1. A population consists of 5 numbers 3, 6, 9, 15 and 27. Consider all possible samples of size 3 that can be drawn without replacement from this population. Find
(i) The population mean (ii) The population standard deviation
(iii) The mean of the sampling distribution of the means
(iv) The standard deviation of the sampling distribution of the means.

Sol:- Given, population : 3, 6, 9, 15, 27

Population size (N) : 5

Sample size (n) : 3 (drawn without replacement)

(i) The population mean (μ)

$$\mu = \frac{\Sigma x_i}{N} = \frac{3+6+9+15+27}{5} = \frac{60}{5} = 12$$

(ii) The population standard deviation (σ)

$$\sigma = \sqrt{\frac{\Sigma (x_i-\mu)^2}{N}} = \sqrt{\frac{(3-12)^2+(6-12)^2+(9-12)^2+(15-12)^2+(27-12)^2}{5}}$$

$$= \sqrt{\frac{(-9)^2+(-6)^2+(-3)^2+(3)^2+(15)^2}{5}}$$

$$= \sqrt{\frac{81+36+9+9+225}{5}} = \sqrt{\frac{360}{5}} = \sqrt{72} \approx 8.485$$

(iii) The mean of the sampling distribution of the means

By statistical theory, the mean of the sampling distribution of means ($\mu_{\bar{x}}$) is always equal to the population mean (μ):

$$\mu_{\bar{x}} = \mu = 12$$

(iv) The standard deviation of sample distribution of Means

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} \sqrt{\frac{N-n}{N-1}} = \frac{\sqrt{72}}{\sqrt{3}} \sqrt{\frac{5-3}{5-1}} = \sqrt{24}\,\frac{1}{\sqrt{2}}$$

$$\sigma_{\bar{x}} = \sqrt{12} \approx 3.464$$

2. A population consists of numbers 2, 3, 6, 8 and 11. Consider all possible samples of size 2 that can be drawn with replacement from this population. Find i) The population mean (ii) The population standard deviation (iii) the mean of the sampling distribution of the means (iv) The standard deviation of the sampling distribution of the means.

Sol:- Given data, Population : 2, 3, 6, 8, 11

Population size (N) : 5

Sample size (n) : 2 (drawn with replacement)

(i)

$$\text{Population Mean } (\mu) = \frac{\Sigma x_i}{N} = \frac{2+3+6+8+11}{5} = \frac{30}{5} = 6$$

$$\boxed{\mu = 6}$$

(ii)

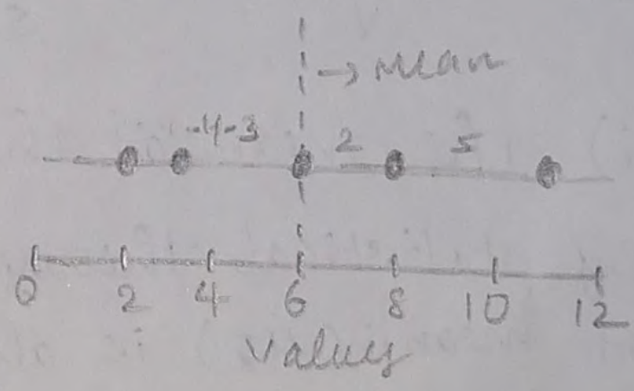
$$\text{standard deviation } (\sigma) = \sqrt{\frac{\Sigma (x-\mu)^2}{N}} =$$

$$= \sqrt{\frac{(2-6)^2 + (3-6)^2 + (6-6)^2 + (8-6)^2 + (11-6)^2}{5}}$$

$$= \sqrt{\frac{(-4)^2 + (-3)^2 + (0)^2 + (2)^2 + (5)^2}{5}} = \sqrt{\frac{16+9+4+25}{5}}$$

$$= \sqrt{\frac{54}{5}} = \sqrt{10.8} \approx 3.2863$$

$$\boxed{\sigma \approx 3.2863}$$



(iii) when sampling with replacement, the total no. of possible samples of size $n=2$ is $N^n = 5^2 = 25$.

By statistical theory, the mean of the sampling distribution of means ($\mu_{\bar{x}}$) is always exactly equal to the population mean

$$\mu_{\bar{x}} = \mu = 6$$

(iv) For sampling with replacement, the standard error is

$$\sigma_{\bar{x}} = \frac{\sigma}{\sqrt{n}} = \frac{\sqrt{10.8}}{\sqrt{2}} = \sqrt{5.4} \approx 2.3238$$

3. Find the median for the following table

Group	50-54	55-59	60-64	65-69	70-74	75-79
Frequency	4	8	10	9	7	6

Sol:- create a continuous frequency table

original Group	Continuous class Interval	Frequency (f)	cumulative frequency (cf)
50-54	49.5-54.5	4	4
55-59	54.5-59.5	8	12
60-64	59.5-64.5	10	22 (Median class)
65-69	64.5-69.5	9	31
70-74	70.5-74.5	7	38
75-79	74.5-79.5	6	44

Total frequency $(N) = 44 \quad \Rightarrow N/2 = 44/2 = 22$

looking at the cf column, the value 22 falls exactly at the end of the 59.5-64.5. so this our median class.

$$\Rightarrow \text{Median} = L + \left[\frac{N/2 - cf}{f}\right] \times h$$

$$= 59.5 + \left[\frac{22 - 12}{10}\right] \times 5 = 59.5 + 5 = 64.5$$

where cf → cumulative frequency of the preceding class

f → frequency of median class

h → class width of median class

L → Lower boundary of median class

Median = 64.5

4. What are the types of data? Explain the methods of collection of data.

Sol:- * Types of data

I. classification by source

1. Primary Data: Original data that is collected firsthand by a researcher specifically for the research problem at hand. It is raw, unique, & directly addresses the research objective.

2. Secondary Data :- Data that has already been collected, compiled and published by someone else for another purpose

II. classification by Nature

1. Quantitative Data (Numerical): Data expressed in numbers that can be measured or counted.

→ Discrete data: Countable whole values
(e.g., no. of stu in a class)

→ Continuous data: Measurable values that can take any value within a range (e.g., height, temp, time).

2. Qualitative Data (Descriptive): Non-numerical data that describes qualities, characteristics, or attributes
(e.g., eye colour, opinions)

* Methods of Data collection

Primary data :- primary methods involve direct interaction with the subject or environment to gather fresh insights

→ Observation Method

- Gathering data by directly watching and recording behaviours, events or physical characteristics without interacting with the subjects.

Ex:- Counting the no. of cars that pass through an intersection to analyze traffic flow.

→ Interview Method

- A direct face to face or verbal communication channel where the interviewer asks questions. It can be structured or unstructured

Ex:- A researcher interviewing focus grps or field experts to understand Consumer habits

3. Questionnaire / Survey Method
   - A set of printed or digital questions delivered to a sample population.

   Ex:- Distributing a google form

4. Experimental Method
   - A scientific approach where the researcher manipulates 1 or more independent variables to observe the effect on a dependent variable under controlled continuous conditions.

Secondary Data collection Methods

↓

It rely on extracting already existing information from established databases.

→ Internal sources - Records within an organization, such as sales invoices or past reports.

→ External sources - Govt. publications, syndicated services, Web / Media sources.

5. A random sample of size 100 is taken from an infinite population having the mean $\mu = 76$ and the variance $\sigma^2 = 256$. Find the probability that will be (i) lies between 75 and 78. (ii) More than 72.

Sol:- Given data, Sample size (n) = 100

Population mean = $\mu = 76$

Population Variance (σ^2) = 256

Population standard deviation (σ) = $\sqrt{256} = 16$

According central limit theorem,

Mean ($\mu_{\bar{x}}$) = $\mu = 76$

standard error ($\sigma_{\bar{x}}$) = $\frac{\sigma}{\sqrt{n}} = \frac{16}{\sqrt{100}} = \frac{16}{10} = 1.6$

The standard normal variable z is calculated using the formula:

$$z = \frac{\bar{x} - \mu_{\bar{x}}}{\sigma_{\bar{x}}} = \frac{\bar{x} - 76}{1.6}$$

(i) we need to find $P(75 \leq \bar{x} \leq 78)$

Convert $\bar{x} = 75$ to a z-score

$$z_1 = \frac{75-76}{1.6} = \frac{-1}{1.6} = -0.625$$

Convert $\bar{x} = 78$ to a z-score

$$z_2 = \frac{78-76}{1.6} = \frac{2}{1.6} = 1.25$$

$$P(-0.625 \leq z \leq 1.25) = P(z \leq 1.25) - P(z \leq -0.625)$$

by using standard normal distribution table values:

$$\Phi(1.25) \approx 0.8944$$

$$\Phi(-0.625) \approx 0.2660$$

$$P(-0.625 \leq z \leq 1.25) = 0.8944 - 0.2660 = 0.6284$$

(ii) we need to find $P(\bar{x} > 72)$

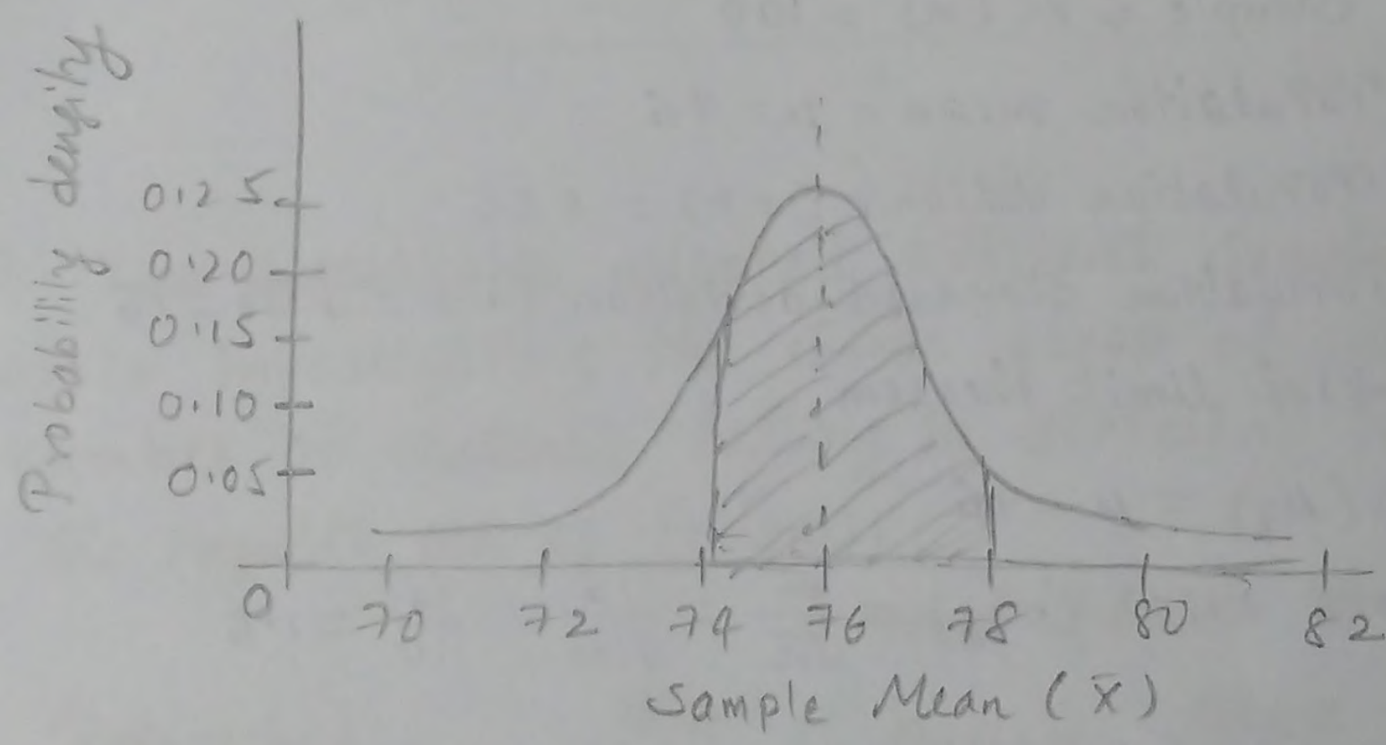
Convert $\bar{x} = 72$ to a z-score:

$$z = \frac{72-76}{1.6} = \frac{-4}{1.6} = -2.5$$

Calculate the probability:

$$P(z > -2.5) = 1 - P(z \leq -2.5)$$

$$= 1 - 0.0062 = \underline{0.9938}$$



6. Find 95% Confidence limits for the mean of a normally distributed population from which the following sample was taken 15, 17, 10, 18, 16, 9, 7, 11, 13, 14, 20, 13, 19, 25, 18, 16.

Sol:- → calculate the Sample Mean

Sample data Contains $n = 16$ values:

15, 17, 10, 18, 16, 9, 7, 11, 13, 14, 20, 13, 19, 25, 18, 16

$$\Sigma x = 15+17+10+18+16+9+7+11+13+14+20+13+19+25+18+16 = 241$$

$$\bar{x} = \frac{\Sigma x}{n} = \frac{241}{16} = 15.0625$$

→ calculate the Sample standard deviation (s)

$$s^2 = \frac{\Sigma (x-\bar{x})^2}{n-1}$$

x	$(x-\bar{x})$	$(x-\bar{x})^2$
15	−0.0625	0.0039
17	1.9375	3.7539
10	−5.0625	25.6289
18	2.9375	8.6289
16	0.9375	0.8789
9	−6.0625	36.7539
7	−8.0625	65.0039
11	−4.0625	16.5039
13	−2.0625	4.2539
14	−1.0625	1.1289
20	4.9375	24.3789
13	−2.0625	4.2539
19	3.9375	15.5039
25	9.9375	98.7539
18	2.9375	8.6289
16	0.9375	0.8789
Sum		314.9375

$$s^2 = \frac{314.9375}{16-1} = \frac{314.9375}{15} \approx 20.9958 \Rightarrow s = \sqrt{20.9958} \approx 4.5821$$

⇒ Critical Value ($t_{\alpha/2}$)

- Degrees of freedom (df) : $n-1 = 16-1 = 15$
- Significance level (α) : $1-0.95 = 0.05$

$\Rightarrow \alpha/2 = 0.025$

→ from the student's t-distribution table for df
= 15 at a two-tailed 95% confidence level:

$$t_{0.025,15} = 2.131$$

⇒ Compute the confidence limits

$$CI = \bar{x} \pm t_{\alpha/2}\left(s/\sqrt{n}\right)$$

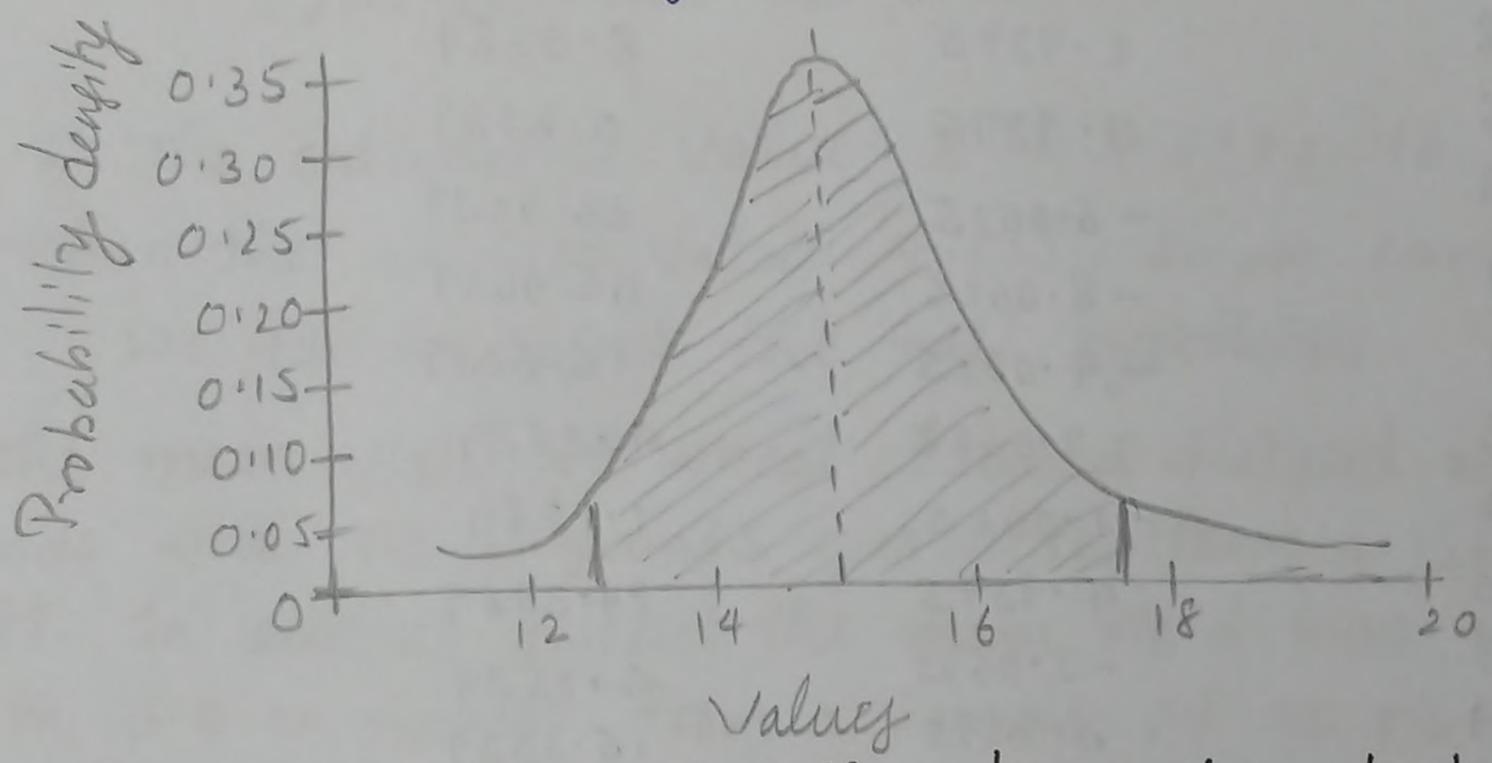
$$SE = \frac{4.5821}{\sqrt{16}} = \frac{4.5821}{4} = 1.1455$$

$$E = 2.131 \times 1.1455 \approx 2.4411$$

Lower limit : $15.0625 - 2.4411 = 12.6214$

upper limit : $15.0625 + 2.4411 = 17.5036$

⇒ The 95% confidence limits for the population mean all approximately (12.62, 17.50)



7. Two researchers adopted different sampling techniques while investigating the same group of students to find the number of students falling in different intelligence levels. The results are as follows:

Researcher	No. of students in each level				
	Below Average	Average	Above Average	Genius	Total
X	86	60	44	10	200
Y	40	33	25	2	100
Total	126	93	69	12	300

would say that the Sampling techniques adopted by the two researchers are significantly different?

Sol:- Null Hypothesis (H_0): There is no significant difference between the Sampling techniques adopted by the two researchers.

Alternative Hypothesis (H_1): There is no significant diff between the sampling techniques adopted by the two researchers.

→ calculate the expected frequencies

$$E = \frac{\text{Row total} \times \text{column total}}{\text{grand total}}$$

grand total (N) = 300

Researches	Below Avg	Avg	Above Avg	Genius	Total
X (Expected)	$\frac{200\times126}{300} = 84$	$\frac{200\times93}{300} = 62$	$\frac{200\times69}{30} = 46$	$\frac{200\times12}{300} = 8$	200
Y (expected)	$\frac{100\times126}{300} = 42$	$\frac{100\times93}{300} = 31$	$\frac{100\times69}{300} = 23$	$\frac{100\times12}{300} = 4$	100
Total	123	93	69	12	300

According to, we need find chi-square (χ^2) statistic

$$\chi^2 = \sum \frac{(O-E)^2}{E}$$

O → observed frequency

E → Expected frequency

category	observed (O)	Expected (E)	(O-E)	$\frac{(O-E)^2}{E}$
$X_{(Below\ Avg)}$	86	84	2	$\frac{(2)^2}{84} = 0.0476$
$X_{(Avg)}$	60	62	-2	$\frac{(-2)^2}{62} \approx 0.0645$
$X_{(Above\ Avg)}$	44	46	-2	$\frac{(-2)^2}{46} \approx 0.0870$
$X_{(Genius)}$	10	8	2	$\frac{(2)^2}{8} \approx 0.3200$
$Y_{(Below\ Avg)}$	40	42	-2	$\frac{(-2)^2}{42} \approx 0.0952$
$Y_{(Avg)}$	33	31	2	$(2)^2/31 \approx 0.1290$

$Y_{(Above\ Avg)}$	25	23	2	$(2)^2/23 \approx 0.1739$
$Y_{(Genius)}$	2	4	-2	$(-2)^2/4 = 1.000$

$$\chi^2 = 0.0476 + 0.0645 + 0.0870 + 0.5000 + 0.0952 + 0.1290 + 0.1739 + 1.0 = 2.0972$$

$$\boxed{\chi^2 = 2.0972}$$

Now find Degrees of freedom & critical value

$$df = (Rows - 1) \times (columns - 1) = (2-1) \times (4-1) = 3$$

→ by using a standard chi-square distribution table at a typical 5% significance level ($\alpha = 0.05$)

Critical value for df=3 is 7.815

∴ The calculated value $\chi^2 = 2.0972$ is much less than the critical value 7.815, so we conclude that we fail to reject the null hypothesis.

8. The mean yield of wheat from a district was 210 Pounds with S.D 10 pounds per acre from a Sample of 100 plots. In another district the mean yield was 220 pounds with S.D 12 pounds from a sample of 50 plots. Test whether there is any significant difference between the mean yields of crops in the two districts.

Sol:- Given data,

→ District 1:

- Sample mean ($\bar{x}_1$) = 210 pounds
- standard deviation (σ_1) = 10 pounds
- Sample size (n_1) = 100 plots

→ District 2:

- Sample mean ($\bar{x}_2$) = 220 pounds
- standard deviation (σ_2) = 12 pounds
- Sample size (n_2) = 50 plots

→ we need to compute test statistic (z-score)

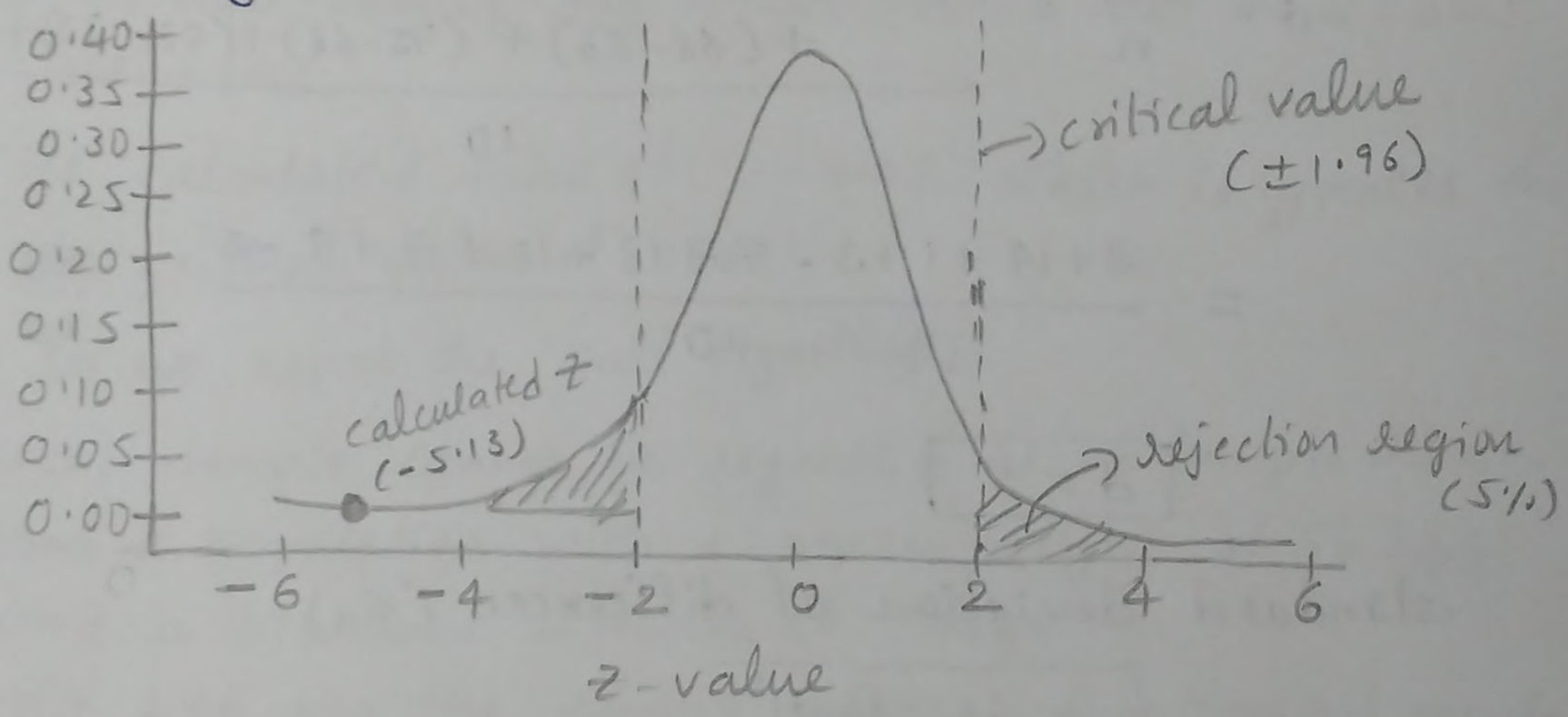
Since both sample sizes are large ($n_1, n_2 \geq 30$)
we use standard formula for a two sample z-test:

$$z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} = \frac{210 - 220}{\sqrt{\frac{10^2}{100} + \frac{12^2}{50}}}$$

$$= \frac{-10}{\sqrt{\frac{100}{100} + \frac{144}{50}}} = \frac{-10}{\sqrt{1 + 2.8}} = \frac{-10}{\sqrt{3.8}} = \frac{-10}{1.949}$$

$$z \approx -5.13$$

Taking absolute value. $|z| = 5.13$



→ At a standard 5% significance level ($\alpha = 0.05$)
the critical value for a two-tailed test is ±1.96.

→ Since, our calculated z is 5.13 is much greater than critical value 1.96.

Therefore, we reject the null hypothesis.

→ There is highly significant evidence to conclude that the mean yields of crops in the two districts are different.

9. Ten soldiers participated in a shooting competition in the first week. After intensive training they participated in the competition in the second week. Their scores before and after training are given as follows:

Scores Before	67	24	57	55	63	54	56	68	33	43
Scores After	70	38	58	58	56	67	68	75	42	38

Do the data indicates that the soldiers have been benefited by the training?

Sol:- From data, No. of pairs $(n) = 10$

Mean of differences $(\bar{d})$

$$\bar{d} = \frac{\Sigma d}{n} = \frac{(70-67)+(38-24)+(58-57)+(56-63)+(67-54)+(68-56)+(75-68)+(42-33)+(38-43)}{10}$$

$$= \frac{3+14+1+3-7+18+12+7+9-5}{10} = \frac{50}{10} = 5$$

$$\boxed{\bar{d} = 5}$$

standard deviation of differences (s_d)

$$s_d = \sqrt{\frac{\Sigma d^2 - \frac{(\Sigma d)^2}{n}}{n-1}}$$

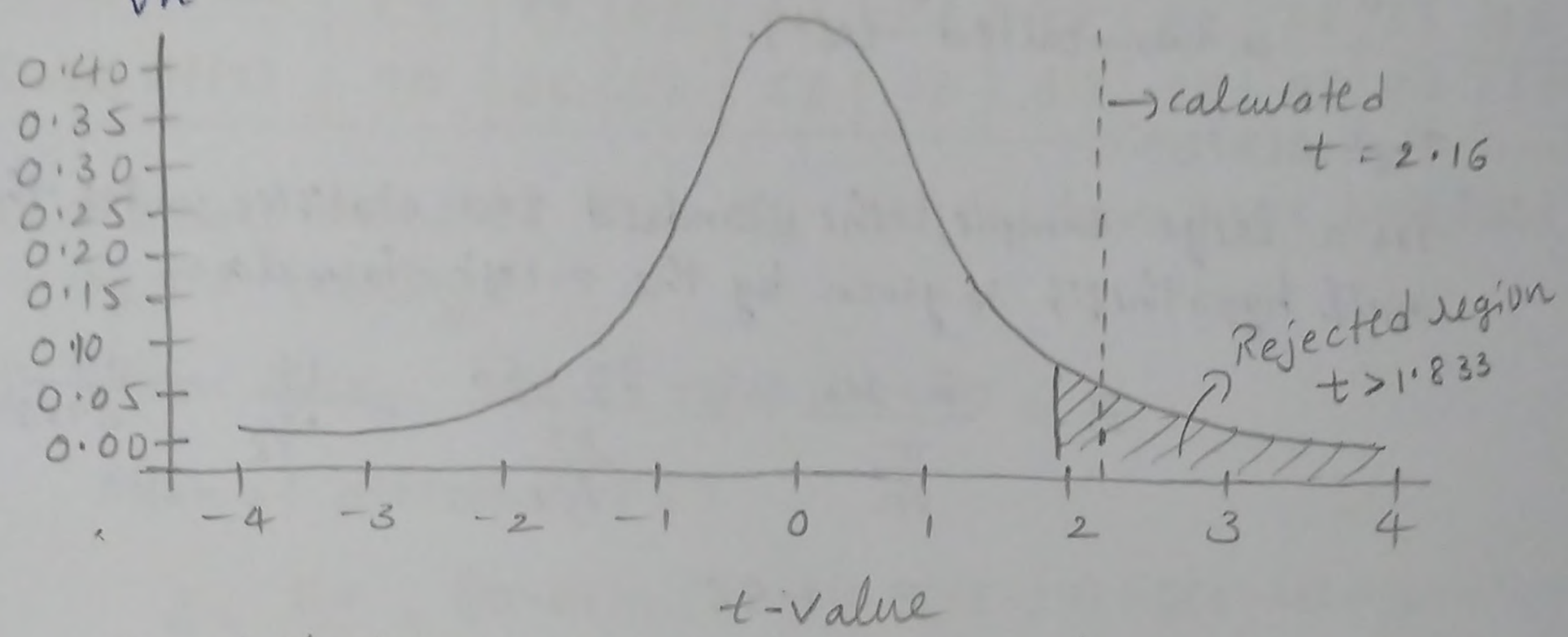
$$= \sqrt{\frac{(3)^2+(14)^2+(1)^2+(+3)^2+(-7)^2+(13)^2+(12)^2+7^2+(9)^2+(-5)^2-\frac{(50)^2}{10}}{10-1}}$$

$$= \sqrt{\frac{9+196+1+9+49+169+144+49+81+25-250}{9}}$$

$$= \sqrt{\frac{482}{9}} = \sqrt{53.556} \approx 7.318$$

→ we need to calculate test statistic (t)

$$t = \frac{\bar{d}}{\frac{S_d}{\sqrt{n}}} = \frac{S}{\frac{7.318}{\sqrt{10}}} = \frac{S}{\frac{7.318}{3.162}} = \frac{S}{2.314} \approx 2.16$$



• One tailed Paired t-test at 5% Significance level

→ degrees of freedom (df) = n-1 = 9

→ At a standard 5% level of significance ($\alpha = 0.05$) for a one tailed test with 9 degrees of freedom, the table value is 1.833.

Since our calculated value t = 2.16 is greater than the critical value ($t_{critical} = 1.833$).

So we reject the null hypothesis.

Yes, the data indicates that the soldiers have significantly benefited from the training at the 5% significance level.

10. A sample of 64 students have a mean weight of 70kgs. Can this be regarded as a sample from a population with mean weight 56 kgs and standard deviation 25 kgs?

Sol:- Given data,

Sample size (n) = 64

Sample mean ($\bar{x}$) = 70 kgs

Sample standard deviation (S) = 25 kgs

Hypothesized population mean (μ) = 56 kgs

→ Null Hypothesis (H_0): $\mu = 56$ kgs (The sample can be regarded as coming from a population with a mean weight of 56 kgs)

→ Alternative Hypothesis (H_1): $\mu \neq 56$ kgs (The sample cannot be regarded as coming from this population; this indicates a two-tailed test).

Test statistic

for a large sample, the standard test statistic under the null hypothesis is given by the z-test formula:

$$Z = \frac{\bar{x} - \mu_0}{\frac{\sigma}{\sqrt{n}}} = \frac{70 - 56}{\frac{25}{\sqrt{64}}} = \frac{14}{25/8} = \frac{14}{3.125}$$

$$Z = 4.48$$

→ The critical values for a two-tailed test at $\alpha = 0.05$ are ± 1.96.

→ The calculated value of $Z = 4.48$ which is greater than the critical value of 1.96.

So we reject the Null Hypothesis.

→ Hence, Sample cannot be regarded as a Sample drawn from a population with a mean weight of 56 kgs and a standard deviation of 25kgs.

→ The diff b/w the Sample mean & hypothesized population mean if highly Significant.

11. Fit a Parabola to the following data

x	0	1	2	3	4
y	1	1.8	3.3	4.5	6.3

Sol:- For a Parabola $y = a + bx + cx^2$ with n data points, the three normal equations are:

1. $\Sigma y = n.a + b\Sigma x + c\Sigma x^2$
2. $\Sigma xy = a\Sigma x + b\Sigma x^2 + c\Sigma x^3$
3. $\Sigma x^2 y = a\Sigma x^2 + b\Sigma x^3 + c\Sigma x^4$

1. $\Sigma y = n \cdot a + b \cdot \Sigma x + c \Sigma x^2$

$n = 5 \rightarrow$ In table we have 5 columns of data

$1 + 1.8 + 3.3 + 4.5 + 6.3 = 5a + b[0+1+2+3+4] + c[(0)^2+(1)^2+(2)^2+(3)^2+(4)^2]$

$16.9 = 5a + 10b + 30c$ — (1)

2. $\Sigma xy = a \Sigma x + b \Sigma x^2 + c \Sigma x^3$

$0(1) + 1(1.8) + 2(3.3) + 3(4.5) + 4(6.3)$

$= a[0+1+2+3+4] + b[0+1+4+9+16] + c[0+1+8+(3)^3+(4)^3]$

$0 + 1.8 + 6.6 + 13.5 + 25.2$

$= a[10] + b[30] + c[100]$

$47.1 = 10a + 30b + 100c$ — (2)

3. $\Sigma x^2 y = a \Sigma x^2 + b \Sigma x^3 + c \Sigma x^4$

$(0)^2(1) + (1)^2(1.8) + (2)^2(3.3) + (3)^2(4.5) + (4)^2(6.3)$

$= a[(0)^2+(1)^2+(2)^2+(3)^2+(4)^2] + b[(0)^3+(1)^3+(2)^3+(3)^3+(4)^3] + c[(0)^4+(1)^4+(2)^4+(3)^4+(4)^4]$

$156.3 = 30a + 100b + 354c$ — (3)

$\Rightarrow$ from eq (1) $a = \frac{16.9 - 10b - 30c}{5} = 3.38 - 2b - 6c$

Substitute a in eq (2)

$10(3.38 - 2b - 6c) + 30b + 100c = 47.1$

$33.8 - 20b - 60c + 30b + 100c = 47.1$

$33.8 + 10b + 40c = 47.1$

$10b + 40c = 13.3$ — (4)

Substitute a into Eq (3)

$30(3.38 - 2b - 6c) + 100b + 354c = 156.3$

$101.4 - 60b - 180c + 100b + 354c = 156.3$

$40b + 174c = 54.9 \rightarrow$ Eq (5)

Multiply eq (4) by 4 & subtract it from eq (5)

$(40b + 174c) - (40b + 160c) = 54.9 - 53.2$

$$14C = 1.7$$

$$C = \frac{17}{140} \approx 0.1214$$

Substitute c back into Equation 4 to find b:

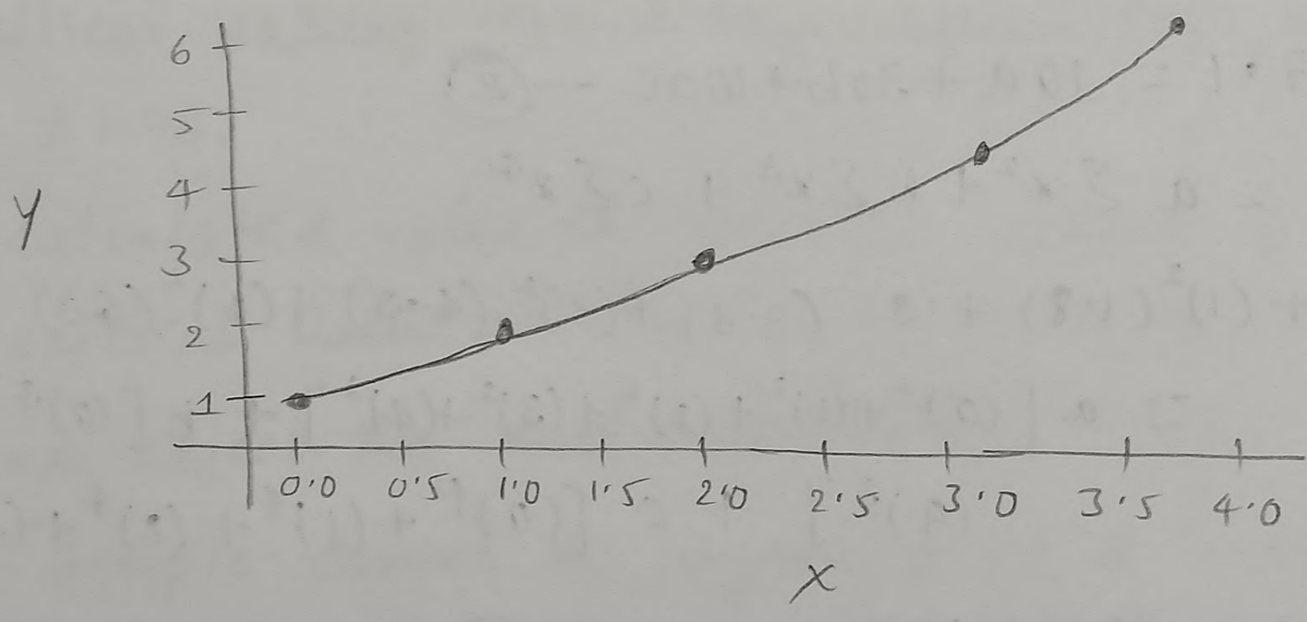
$$10b + 40(0.1214) = 13.3 \Rightarrow 10b = 8.4429 \Rightarrow b = 0.8443$$

Substitute b & c back to find a:

$$a = 3.38 - 2(0.8443) - 6(0.1214) = 0.9629$$

$$y = a + bx + cx^2$$

$$\boxed{y = 0.9629 + 0.8443x + 0.1214x^2}$$



12. For the following data, fit the curve of the form $y = ae^{bx}$.

x	55	56	58	59	60	60	62
y	35	38	38	39	44	43	45

Sol:- Given curve $y = ae^{bx}$

$$\log y = \log a + bx$$

let $Y = \log y$; $\log a = A \Rightarrow a = e^A$; $B = b$

And now $Y = A + Bx$

The normal equations are :-

$$\Sigma Y = nA + B\Sigma x$$

$$\Sigma xY = A\Sigma x + B\Sigma x^2$$

n = no. of observations = 7

$\Rightarrow) \Sigma Y = nA + B\Sigma x \qquad [\because Y = \log y]$

$\log 35 + \log 38 + \log 38 + \log 39 + \log 44 + \log 43 + \log 45$

$= 7A + B[55+58+58+59+60+60+62]$

$3.5553 + 3.6376 + 3.6376 + 3.6636 + 3.7842 + 3.7612 + 3.8067$

$= 7A + 410B$

$25.8462 = 7A + 410B \quad \text{—(1)}$

$\Rightarrow) \quad \Sigma xY = A\Sigma x + B\Sigma x^2$

$55(3.5553) + 56(3.6376) + 58(3.6376) + 59(3.6636) + 60(3.7842)$

$+ 60(3.7612) + 62(3.8067) = A[55+56+58+59+60+60+62]$

$+ B[3025 + 3136 + 3364 + 3481$

$+ 3600 + 3600 + 3844]$

$1515.1197 = 410A + 24050B \quad \text{—(2)}$

from eq (1)

$$A = \frac{25.8462 - 410B}{7}$$

Substitute A in eq (2)

$$1515.1197 = 410\left(\frac{25.8462 - 410B}{7}\right) + 24050B$$

$1515.1197 = 1513.8488 - 24014.2857B + 24050B$

$1.2709 = 35.7143B$

$B \approx 0.0356$

$$A = \frac{25.8462 - 410(0.0356)}{7} \approx 1.6072$$

for original a & b

$b = B = 0.0356$

$a = e^A = e^{1.6072} \approx 4.988$

$\therefore y = 4.9888 e^{0.0356x}$

13. calculate Rank correlation for the following data

x	48	33	40	9	16	16	65	24	16	57
y	13	13	24	6	15	4	20	9	6	19

Sol:- Given

no. of pairs (n) = 10

Ties in x : The value 16 appears 3 times ($m_1 = 3$)

Ties in y : The value 13 appears 2 times ($m_2 = 2$) and 6 appears 2 times ($m_3 = 2$)

Formula for Rank Correlation Coefficient with ties

$$\rho = 1 - \frac{6\left[\Sigma d^2 + \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12}\right]}{n(n^2-1)}$$

x	y	Rank of x (Rx)	Rank of y (Ry)	$d = R_x - R_y$	d^2
48	13	8	5.5	2.5	6.25
33	13	6	5.5	0.5	0.25
40	24	7	10	-3.0	9.00
9	6	1	2.5	-1.5	2.25
16	15	3	7	-4.0	16.00
16	4	3	1	2.0	4.00
65	20	10	9	1.0	1.00
24	9	5	4	1.0	1.00
16	6	3	2.5	0.5	0.25
57	19	9	8	1.0	1.00

$\Sigma d^2 = 41$

$$\rho = 1 - \frac{6\left[41 + \frac{3(3^2-1)}{12} + \frac{2(2^2-1)}{12} + \frac{2(2^2-1)}{12}\right]}{10(10^2-1)} = 1 - \frac{6[41+3]}{10(99)}$$

$$= 1 - \frac{\cancel{6} \times 44}{\cancel{10}(99)_{3}} = 1 - \frac{4}{15} = \frac{11}{15} \approx 0.7333$$

∴ rank correlation coefficient is approx 0.7333, it indicating a strong positive rank correlation b/w x & y.

4. Calculate karl Pearson's coefficient of correlation for the following data

x	12	9	8	10	11	13	7
y	14	8	6	9	11	12	3

Sol:- Given no. of observations (n) = 7

$$\text{Mean of } x\ (\bar{x}) = \frac{\Sigma x}{n} = \frac{12+9+8+10+11+13+7}{7} = \frac{70}{7} = 10$$

$$\text{Mean of } y\ (\bar{y}) = \frac{\Sigma y}{n} = \frac{14+8+6+9+11+12+3}{7} = \frac{63}{7} = 9$$

Correlation formula

$$r = \frac{\Sigma uv}{\sqrt{\Sigma u^2 \cdot \Sigma v^2}} \quad \text{where} \quad u = x-10, \quad v = y-9$$

$$r = \frac{(12-10)(14-9)+(9-10)(8-9)+(8-10)(6-9)+(10-10)(9-9)+(11-10)(11-9)+(13-10)(12-9)+(7-10)(3-9)}{\sqrt{\left[(12-10)^2+(9-10)^2+(8-10)^2+(10-10)^2+(11-10)^2+(13-10)^2+(7-10)^2\right]\times\left[(14-9)^2+(8-9)^2+(6-9)^2+(9-9)^2+(11-9)^2+(12-9)^2+(3-9)^2\right]}}$$

$$= \frac{(2)(5)+(-1)(-1)+(-2)(-3)+(0)(0)+(1)(2)+(3)(3)+(-3)(-6)}{\sqrt{[4+1+4+0+1+9+9][36+9+4+0+9+1+25]}}$$

$$= \frac{10+1+6+0+2+9+18}{\sqrt{28\cdot(84)}} = \frac{46}{\sqrt{2352}} = \frac{46}{48.50} \approx 0.948$$

∴ The karl Pearson coefficient of correlation is 0.948 indicating a strong +ve correlation b/w x & y.

15. The equations of two regression lines are $7x-16y+9=0$ and $5y-4x-3=0$, find the coefficient of correlation and the means of x & y.

Sol:- since the 2 regression lines always intersect at the point of their means $(\bar{x}, \bar{y})$

eq ① → $7x-16y+9=0 \Rightarrow 7x-16y=-9$

eq ② $\Rightarrow -4x+5y=3$

multiply eq ① by 4 & eq ② by 7

$$28x-64y=-36$$
$$-28x+35y=21$$
$$-29y=-15$$

$$y=\frac{15}{29}\approx 0.5172 \Rightarrow \boxed{\bar{y}=0.5172}$$

$$-4x+5\left(\frac{15}{29}\right)=3$$

$$-4x+\frac{75}{29}=3 \Rightarrow -4x=3-\frac{75}{29}=\frac{87-75}{29}=\frac{12}{29}$$

$$\bar{x}=\frac{-3}{29}\approx -0.1034$$

$\boxed{\bar{x}=-0.1034}$

⇒ Coefficient of correlation (r)

$$r^2=b_{yx}+b_{xy}$$

let $7x-16y+9=0$ be regression line of y on x:

$$16y=7x+9 \Rightarrow y=\frac{7}{16}x+\frac{9}{16}$$

regression coefficient $b_{yx}=7/16$

let $-4x+5y-3=0$ be the regression line of x on y

$$x=\frac{5}{4}y-\frac{3}{4} \Rightarrow \text{regression coefficient } b_{xy}=\frac{5}{4}$$

Verification

$$r^2 = b_{yx} + b_{xy} = \frac{7}{16} \times \frac{5}{4} = \frac{35}{64} \approx 0.5469$$

$r^2 \leq 1$, → our assumptions are correct.

both b_{yx} & b_{xy} are +ve.

$$r = \sqrt{\frac{35}{64}} = \frac{\sqrt{35}}{8} = 0.7395$$

$\therefore r = 0.7395$

16. Ten persons are assessed by 3 judges in a beauty Contest. Determine the closeness b/w opinions of judges by rank correlation coefficient.

Score assigned by J-1	65	70	72	76	77	79	72	80	77	72
Score assigned by J-2	59	71	77	71	69	79	69	65	67	80

Sol:-

for J-1 (x)

Positions : 1 2 3 4 5 6 7 8 9 10

Sorted values : 65, 70, 72, 72, 72, 76, 77, 77, 79, 80

ranks (1, 2, $\frac{3+4+5}{3}$ = 4, 6, $\frac{7+8}{2}$ = 7.5, 9, 10)

for J-2 (y)

Positions : 1 2 3 4 5 6 7 8 9 10

Sorted values : 59 65 67 69 69 71 71 77 79 80

ranks : 1 2 3 $\frac{4+5}{2}$ = 4.5, $\frac{6+7}{2}$ = 6.5, 8, 9, 10

x	y	Rank of x (R_x)	Rank of y (R_y)	$d = R_x - R_y$	d^2
65	59	1	1	0	0
70	71	2	6.5	-4.5	20.25
72	77	4	8	-4	16
76	71	6	6.5	-0.5	0.25
77	69	7.5	4.5	3	9
79	79	9	9	0	0

72	69	4	4.5	-0.5	0.25
80	65	10	2	8	64
77	67	7.5	3	4.5	20.25
72	80	4	10	-6	36

$\Sigma d^2 = 166$

⇒ Ties in x : 72 repeats 3 times → $m_1 = 3$ ⇒ $\frac{m_1(m_1^2-1)}{12}$

⇒ $\frac{3^3-3}{12} = 2$

77 repeats 2 times → $m_2 = 2$ ⇒ $\frac{m_2(m_2^2-1)}{12}$

⇒ $\frac{(2)^3-2}{12} = 0.5$

⇒ Ties in y : 69 repeats 2 times → $\frac{2^3-2}{12} = 0.5 \rightarrow m_3 = 2$

71 repeats 2 times → $m_4 = 2 \rightarrow \frac{2^3-2}{12} = 0.5$

$CF = 2 + 0.5 + 0.5 + 0.5 = 3.5$

$CF = \frac{m_1(m_1^2-1)}{12} + \frac{m_2(m_2^2-1)}{12} + \frac{m_3(m_3^2-1)}{12} + \frac{m_4(m_4^2-1)}{12}$

we need to calculate

$$\rho = 1 - \frac{6(\Sigma d^2 + CF)}{n(n^2-1)}$$

$$= 1 - \frac{6(166 + 3.5)}{10(10^2-1)} = 1 - \frac{6 \times 169.5}{10 \times 99}$$

$$= 1 - \frac{1017}{990} = 1 - 1.0273 = -0.0273$$

$$\boxed{\rho = -0.0273}$$

The correlation b/w $\bar{J}-1$ & $\bar{J}-2$ is -0.0273, Indicating very small -ve value, means almost no correlation.